Natural language processing Applications

- Text categorisation
- Text summerisation
- Information extraction
- Sentiment analysis
- Semantic web
- Dialogue syatems
- Machine translation

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Word sense disambiguation

1. Flies like fruits

Time flies like an arrow.

Word flies means different in these two sentences.

2. Monkeys like bananas when they wake up

Monkeys like bananas when they are ripe.

They mean differently in the two sentences

3. I can hear bass sound.

He likes to eat grilled bass.

First one bass is frequency

Second bass is a fish

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TEXT PROCESSING

- Processing raw text
  - Morphological Analysis
    - Running --> run + ing
  - POS Tagging
    - Ram/NNP goes/VB to/TO school/NNP ..
  - Stemming
    - running --> run
  - Parsing
    - Identifying sentence structure
    - S --> NP + VP .

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MAJOR AREAS OF RESEARCH & DEVELOPMENT

- Text Processing
- Morph Analyzer
- POS Tagging
- Parsing
- Machine Translation .....
- Speech Processing
- Text to Speech (TTS)
- Automatic Speech Recognition (ASR)
- Speech to Speech Translation

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TEXT PROCESSING

Machine Translation

Translating content in one natural language to another natural language

Example : Translating and English Sentence to Malaylam with the help of a software.

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Associative Networks

(or Semantic networks)

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Intro

- Used for clustering of pieces of knowledge to semantic groups.
- Pictorial presentation of objects their attributes and the relationship between them and other entities
- Provides a more natural way to map natural language than other representation schemes

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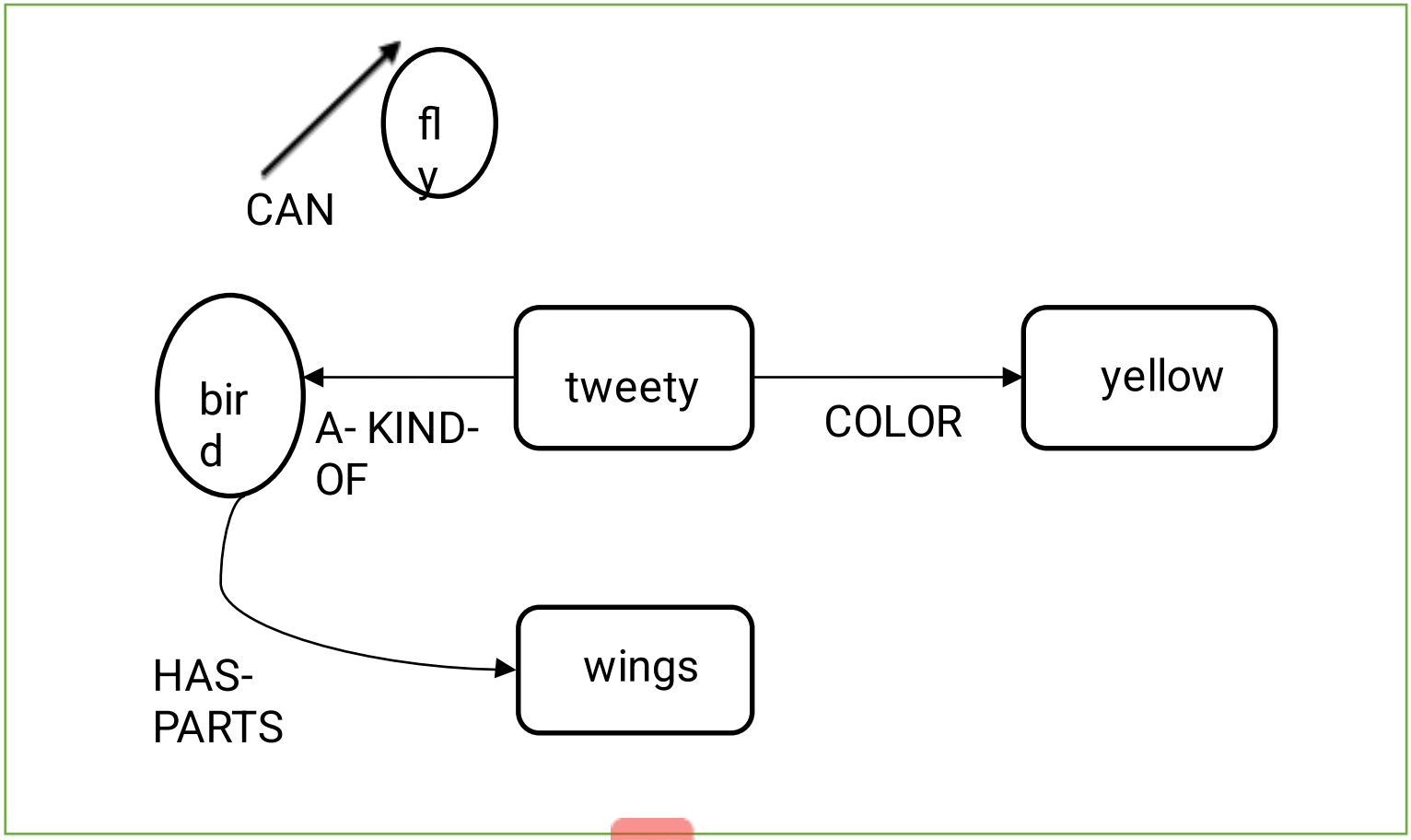
- Associative Networks are directed graphs with labelled nodes arrows and arcs of arrow.
- The language used in constructing a network is based on selected domain primitives for objects relations
  as well as some general primitives.
- A fragment of simple network is illustrated in the figure
- The class has some properties and a specific member of the class named Tweety is shown.



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- Associative networks were introduced by Quillian (1968) to model the semantics of English sentences and words.
- He called these networks semantic networks to specify their intended use.
- He developed a system which “found” the meanings between words by the paths connecting them.
- The knowledge required for the performance of some task is typically contained within a narrow domain or “semantic vicinity” of the task .
- This resembles the way knowledge is stored and retrieved in humans.

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Syntax and Semantics of Associative networks

- Unlike FOPL , there is no generally accepted syntax nor semantics for associative networks.
- Most network systems are based on PL or FOPL with extensions.
- The syntax for any given system is determined by the object and relation primitives chosen and by any special rules used to connect nodes.

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- Basically, the language of associative networks is formed from the letters of the alphabet, both upper and lowercase, relational symbols, set membership and sub sub symbols, decimal digits, square and oval nodes, and directed arcs of arbitrary length. The word symbols used arethose which represent object constants and n-ary relation constants.
- Nodes are commonly used for objects or nouns, and arcs for relations
- The direction of an arc is usually taken from the first to the subsequent arguments as they appear in a relational statement.

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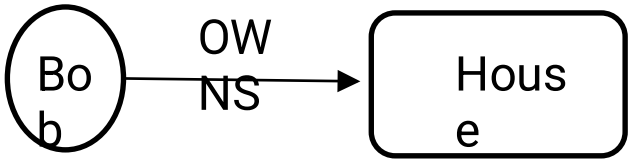
Thus OWNS(bob,house) would be written as:



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- A number of arc relations have become common among users. They include predicates as ISA , MEMBER-OF , SUBSET-OF , AKO (a-kind-of), HAS PARTS , INSTANCE-OF , AGENT , ATTRIBUTES , SHAPED-LIKE and so forth . Less common arcs have also been used to express modality relations , linguistics case relations , quantifiers, set relations etc
- One particular arc or link , the ISA link has taken on a special meaning . It signifies that Bob is a professor and that the state university system in an institute of higher learning.

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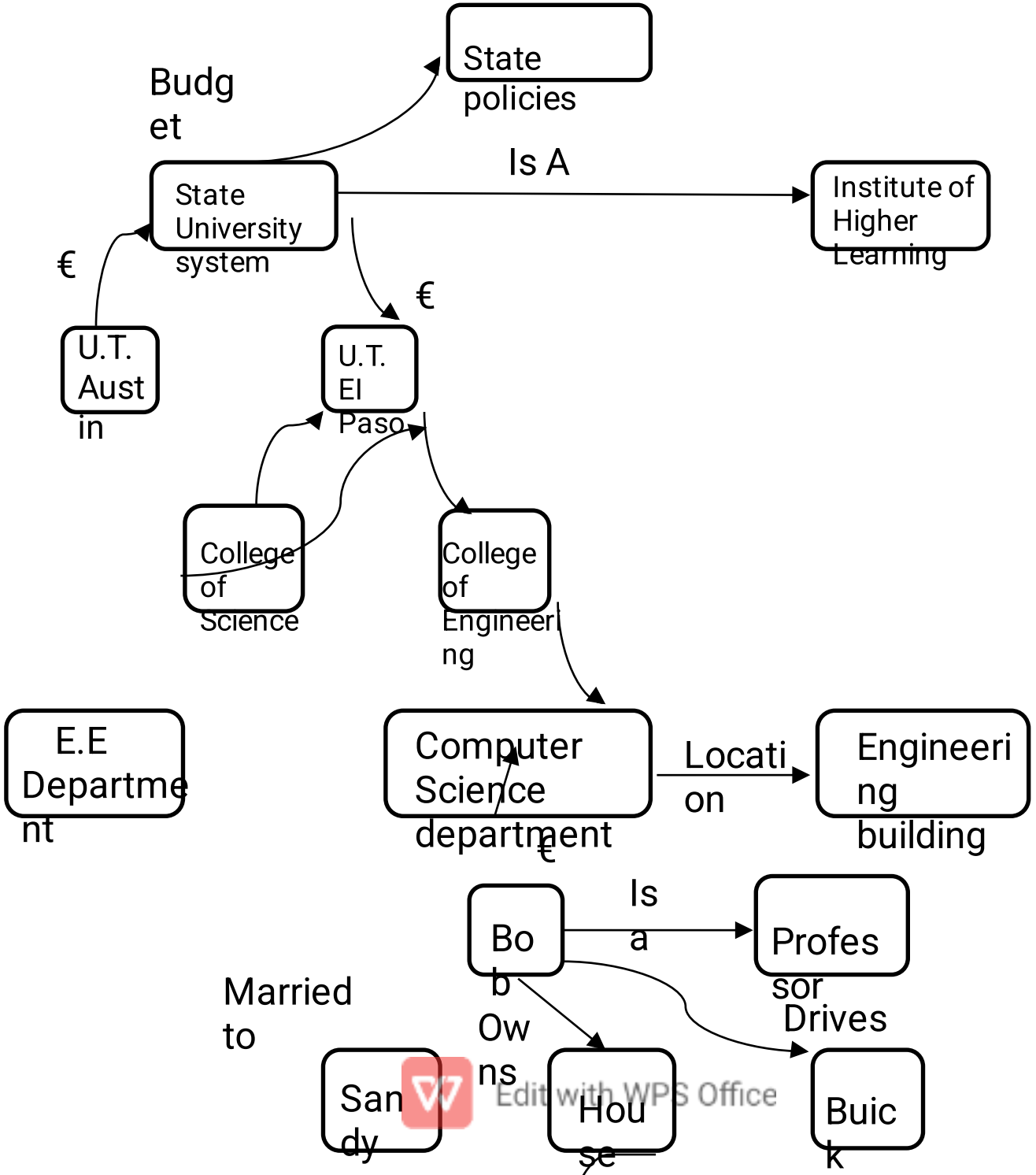

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- It helps in inheriting the properties of higher class, called property inheritance. All properties common to higher class and sub class are taken to the higher level. Special property for the subclass only is attached to it.
- **Property inheritance:**
- To retrieve a value V for attribute A of an object O:
- 1. Find O in the knowledge base
- 2. If there is a value there for attribute A , report that value
- 3. Otherwise, see if there is value for the attribute instance, otherwise report fail.
- 4. Otherwise move to the node corresponding to that value and look for a value for the attribute A. If one is found report it
- 5.Otherwise, do until there is no value for the isa attribute or until the answer is found:
- a. get the value of the isa attribute and move to that node
- b. see if there is a value for the attribute A . If there is report it.

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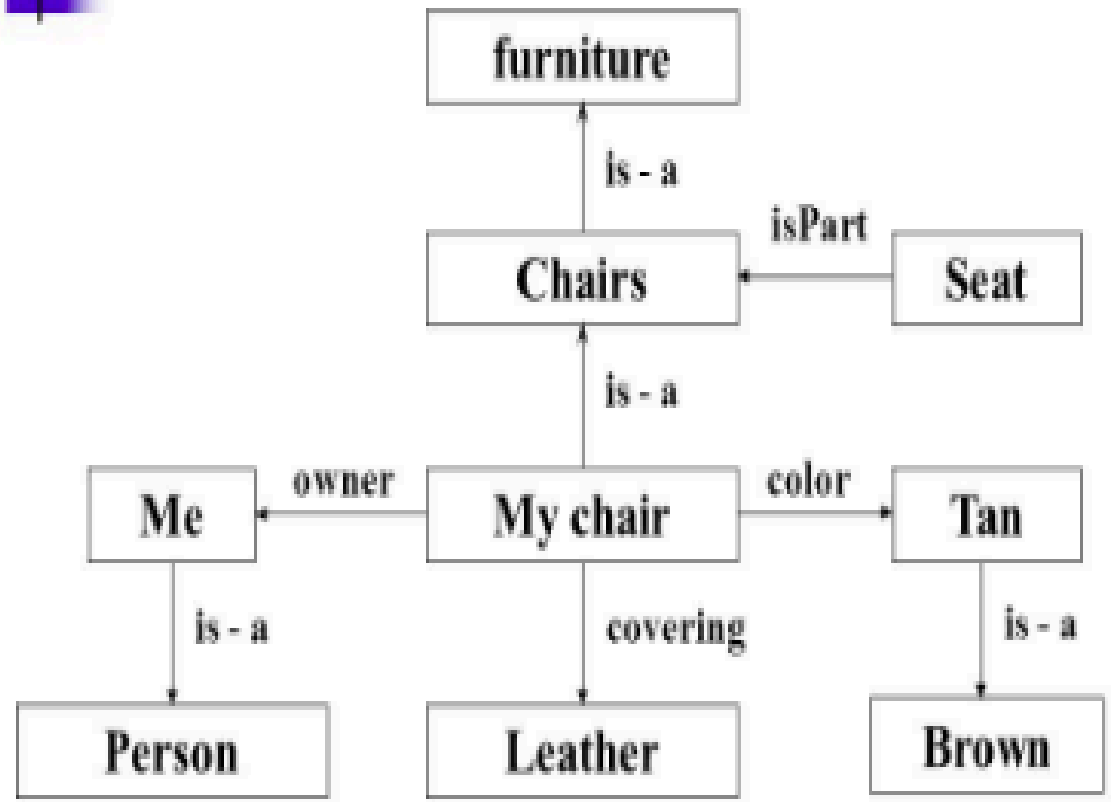
Semantic Networks

Relations



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Steps in natural language processing

- Morphological Analysis
- Syntactic Analysis
- Semantic Analysis
- Discourse Analysis
- Pragmatic Analysis

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- **Morphological analysis**
- Individual words are analysed into their components. Nonword tokens and punctuations are separated from words
- e.g:- carried=carry + ed (tense suffix)
- Sings = sing + s (singular verb suffix)
-
- Root words are transformed to different grammatical categories like noun, verb, adjective, adverb etc.

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- **Syntactic analysis**
- Grammatical structure of sentence is analysed
- Some woed sequence may be rejected if it doesnot follow the rules of the language
- “ boy the go the to store” will be rejected
- **Semantic analysis**
- Determines possible mening of sentences. Sentences which have no meaning are rejected
- e.g:- “colorless gree ideas sleep furiously” has no meaning

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Simple grammar/ transformational grammar

S -> NP VP

NP -> the NP1

NP -> PRO | PN | NP1

NP1 -> ADJS N

ADJS -> ε | ADJ ADJS

VP -> V | V NP

N -> file | pointer

PN -> Bill

PRO -> I

ADJ -> short | long | fast

V -> printed | created | want

e.g:- Bill created the short file , Bill printed the file fast

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- **Discourse analysis**
- The meaning of an individual sentence may depend on sentences that precede it and may influence the sentences that may follow it.
- e.g:- “john wanted it”, “it” depends on a prior discourse context
- Also John may influence following sentences
- **Pragmatic analysis**
- The structure representing what was said is reinterpreted to determine what was actualy meant. It finds practically what is to be done.
- e.g:- “do you know what time it is?” should be interpreted as a request to be told time.

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- These five steps may not be independent. They may need to interact for one step to decide on a sentence structure.
- Is the glass jar peanut butter?
- How to group the four nouns: semantic analysis may help the syntactic analyser to find the correct grouping:
- Glass jar and peanut butter
- Possible groupings are: glass, glass jar, glass jar peanut,.....etc

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Simple grammar/ transformational grammar

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ADJS -> ε | ADJ ADJS

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N -> file | pointer

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e.g:- Bill created the short file , Bill printed the file fast

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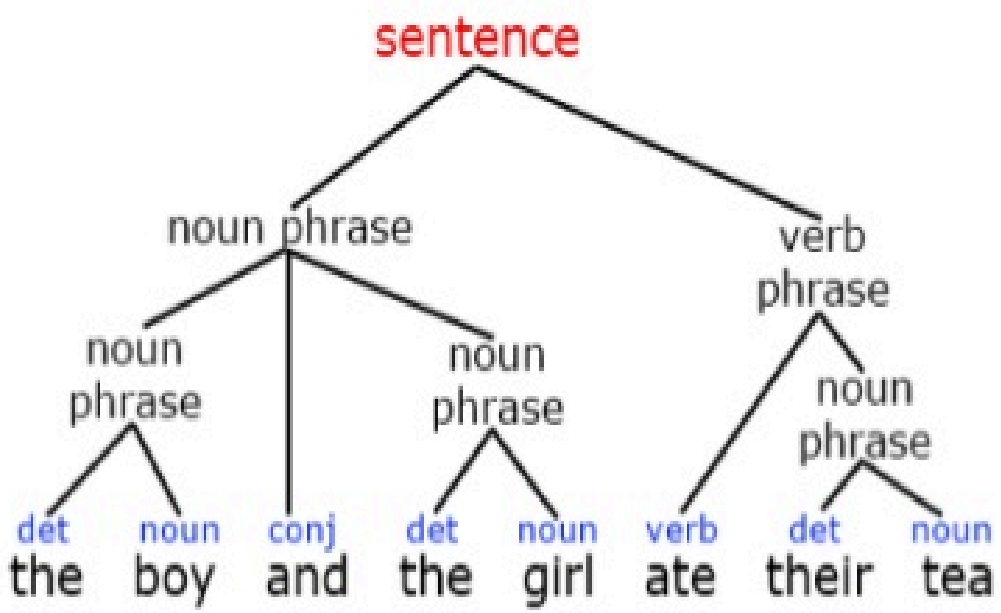
Syntactic Analysis - Grammar



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- The comple steps can be seen with the sample sentence:
- “I want to print Bill’s .init file”
- Morphological analysis
- ‘s is separated from Bill and taken as possessive suffix. .init is classified as file extension
- words are assigned to syntactic categories
- Correct assignment may not be possible for many words like print, file etc. since they can belong to multiple syntactic categories (verb, noun).

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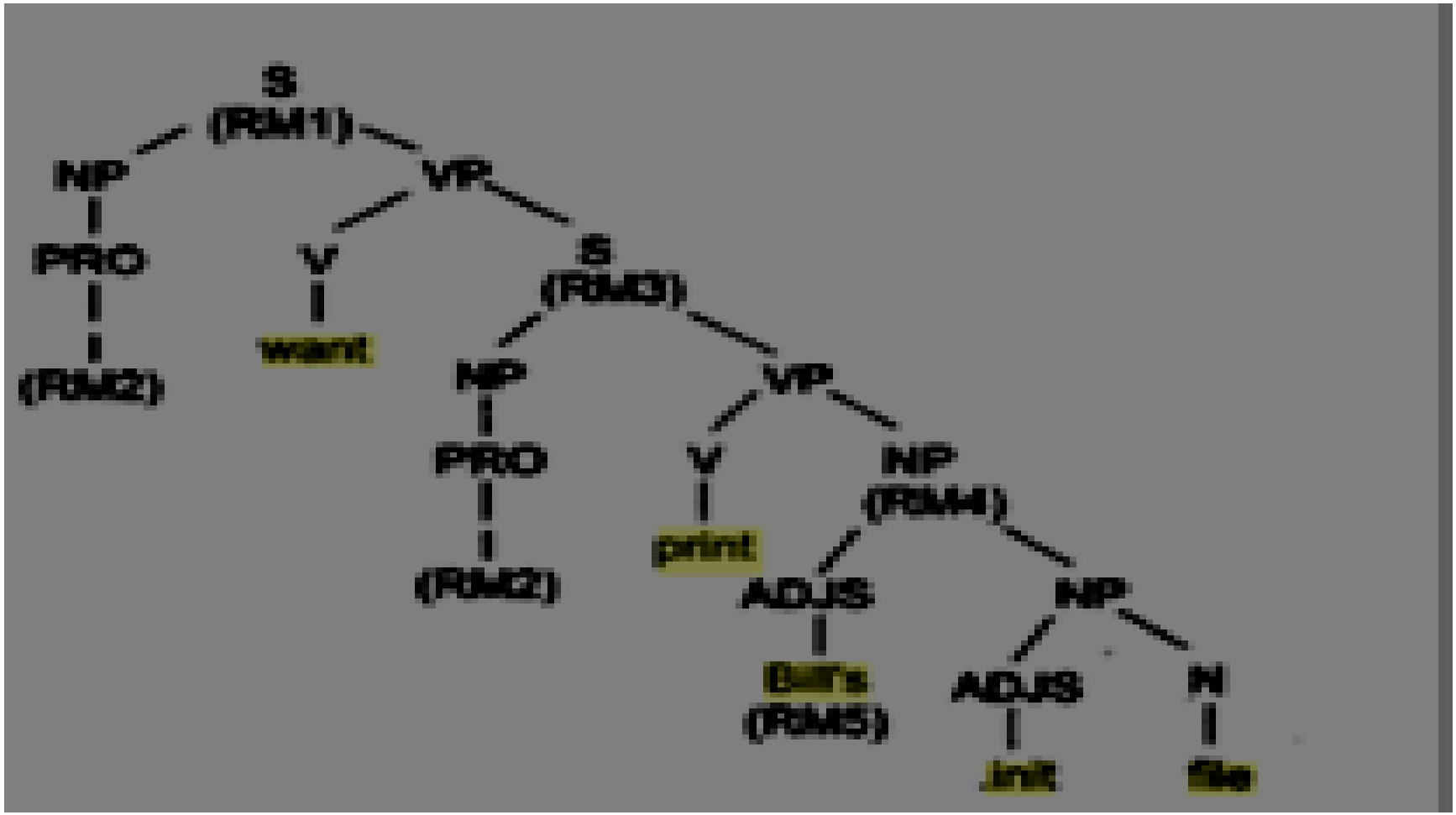
Syntactic analysis



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Semantic analysis

- Semantic analysis needs a semantic network of word meanings in the context.
- It has to assign the words in the sentence to words in the knowledge base
- It also create a correct structure to correspond to the way meanings of the individual words combine with each other.

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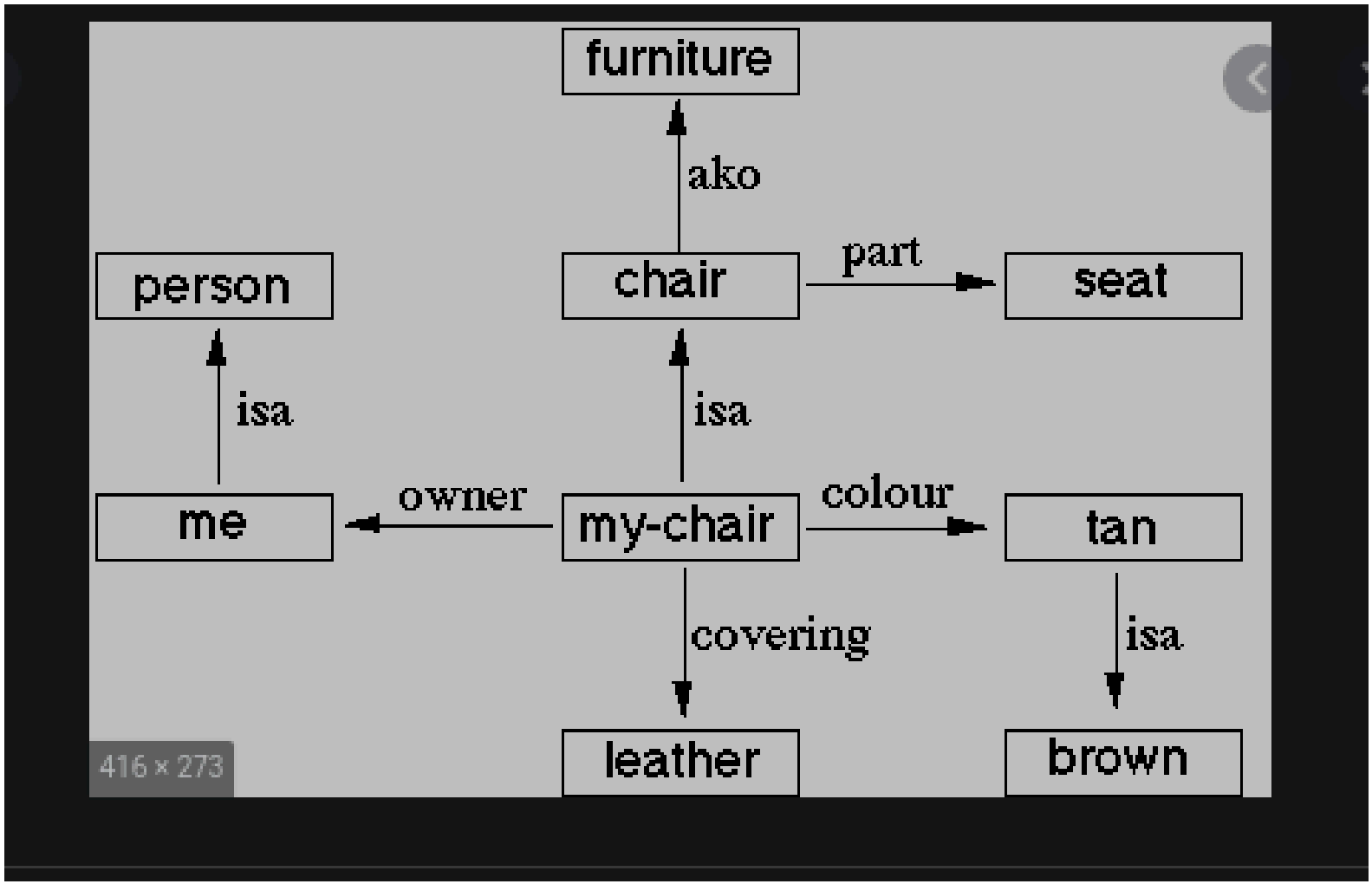
Sample Semantic net

- 

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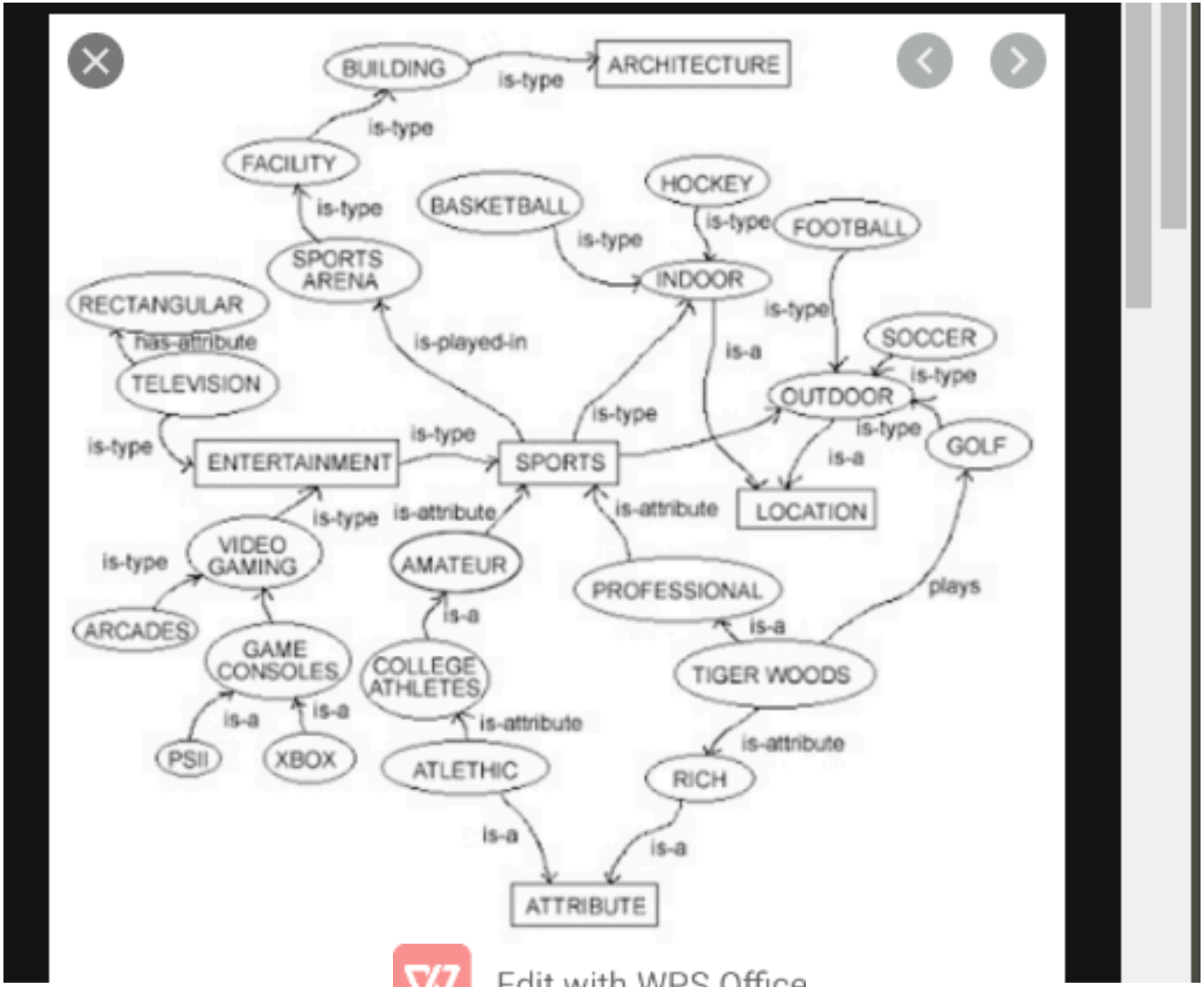
Instance and is-a are marked as is-type - not a good hierarchy

- 

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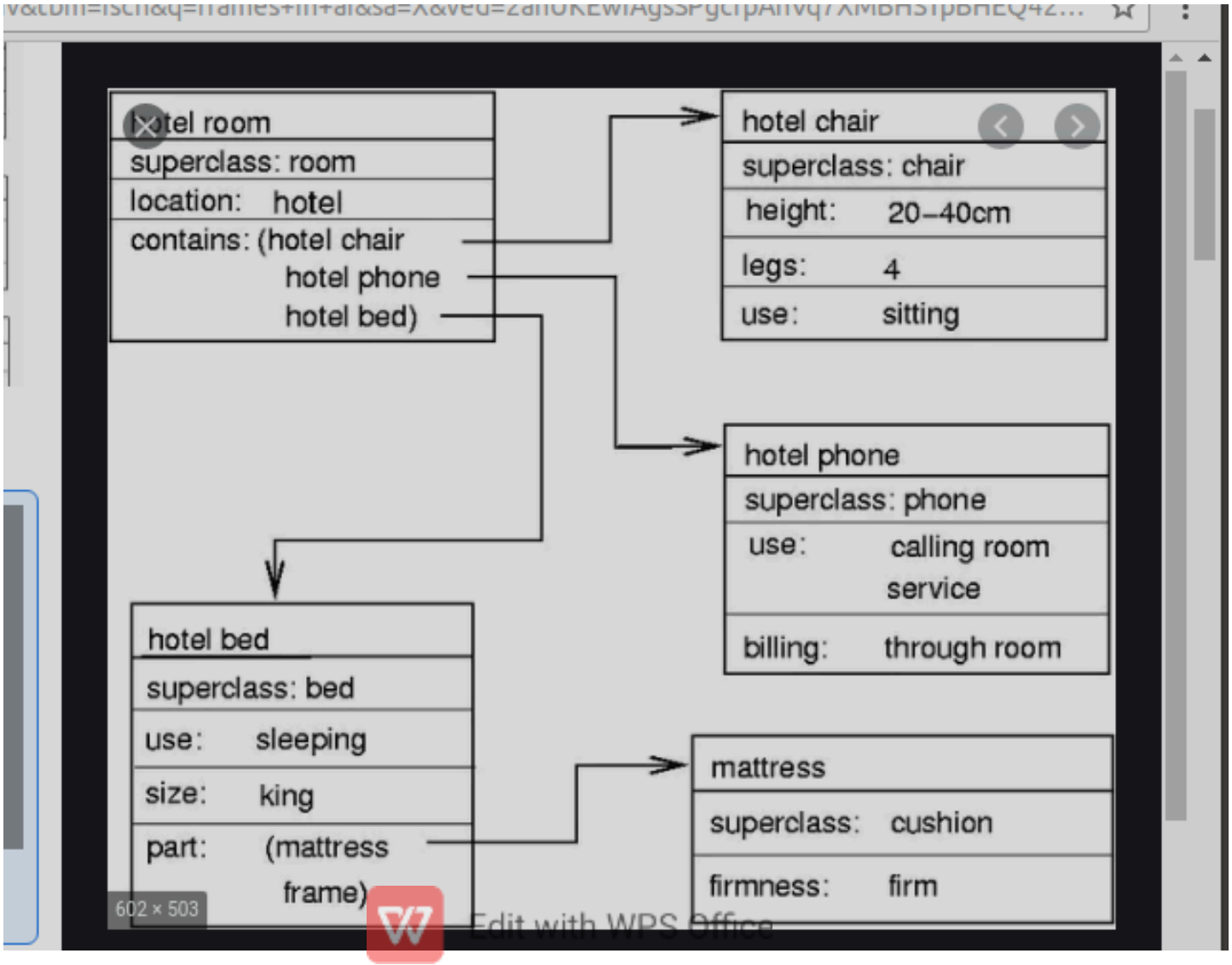
Sample Frames and semantic nets

- 

Frame system to represent the semantics

User

isa :	*Person*
* *login-name* :	must be <string>

User068

instance :	*User*
login-name :	*Susan-Black*

User073

instance :	*User*
login-name :	*Bill-Smith*

F1

instance :	*File-Struct*
name :	stuff
extension :	.init
owner :	*User073*
in-directory :	/wsmith/

File-Struct

isa :	*Information-Object*

Printing

isa :	*Physical-Event*
* *agent* :	must be <animate or program>
* *object* :	must be <information-object>

Wanting

isa :	*Mental-Event*
* *agent* :	must be <animate>
* *object* :	must be <state or event>

Commanding

isa :	*Mental-Event*
* *agent* :	must be <animate>
* *performer* :	must be <animate or program>
* *object* :	must be <event>

This-System

instance :	*Program*

Fig. 15.3 *A Knowledge Base Fragment*

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Partial meaning constructed by semantic analyser

-

RM1		{the whole sentence}
instance :	*Wanting*	
agent :	*RM2*	{I}
object :	*RM3*	{a printing event}
RM2		{I}
RM3		{a printing event}
instance :	*Printing*	
agent :	*RM2*	{I}
object :	*RM4*	{Bill's .init file}
RM4		{Bill's .init file}
instance :	*File-Struct*	
extension :	.init	
owner :	*RM5*	{Bill}
RM5		{Bill}
instance :	*Person*	
first-name :	Bill	

Fig. 15.4 *A Partial Meaning for a Sentence*

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Pragmatic analysis: Final internal structure

Meaning

instance :	*Commanding*
agent :	*User068*
performer :	*This-System*
object :	*P27*

P27

instance :	*Printing*
agent :	*This-System*
object :	*F1*

g. 15.5 *Representing the Intended Meaning*

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- The pragmatic analysis phase constucts the command for printing a file from the structure
- lpr /wsmith/stuff.init

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Case grammar

a form of grammar in which the structure of sentences is analysed in terms of semantic case relationships

This theory analyzes the surface syntactic structure of sentences by studying the combination of deep cases (i.e. semantic roles, such as Agent, Object, Benefactor, Location or Instrument etc.) which are required by a specific verb.

e.g:- the verb "give" requires an Agent (A) and Object (O), and a Beneficiary (B); e.g. "Jones (A) gave money (O) to the school (B).

So john gave mary – is syntactically wrong

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Subject – nominative case
Object – accusative case
Other cases :
receiver- Dative case
location - Locative case

Case relationships are identified with the positions or with the prepositions preceding the nouns

John (subject) gave a flower(object) to Mary(receiver)
Subject and object are identified by their respective positions with respect to verb
The recipient is idntified by the preposition “to”

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Tools

- GATE
- OpenNLP
- NLTK (python)
- Stanford Parser
- Weka for classification

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